The Binomial Theorem

Binomial coefficients

1 Evaluate each binomial coefficient:

a $\binom{6}{2}$ **b** $\binom{8}{0}$ **c** $\binom{5}{5}$ **d** $\binom{10}{3}$

2 Write down row 4 of Pascal's triangle (the row beginning $1, 4, \ldots$) and explain how it relates to the expansion of $(a+b)^4$.

Expansions and specific terms

3 Expand $(x+2)^4$ fully.

4 Find the coefficient of x^3 in the expansion of $(2x-1)^5$.

5 Find the term containing y^4 in the expansion of $(x+y)^7$.

6 Use the binomial theorem to expand $(1+0.02)^5$ to the x^2-style second-order term, and hence estimate 1.02^5 to four decimal places.

More expansions and terms

7 Expand $(2x-y)^3$ fully.

8 Find the middle term of the expansion of $(x+3)^6$ (the term with x^3).

9 Find the constant term in the expansion of $\left(x^2+\frac{1}{x}\right)^6$.

Sums of binomial coefficients

10 Evaluate $\binom{5}{0}+\binom{5}{1}+\binom{5}{2}+\binom{5}{3}+\binom{5}{4}+\binom{5}{5}$ without computing each term, using the identity $\sum_{k=0}^{n}\binom{n}{k}=2^n$.

11 Use the fact that $\binom{n}{k}=\binom{n}{n-k}$ (symmetry of Pascal's triangle) to explain why $\binom{9}{2}=\binom{9}{7}$ without computing either value directly.

Further expansions

12 Find the coefficient of x^4y^3 in the expansion of $(x+2y)^7$.

13 Find the first three terms in the expansion of $(1-2x)^6$ in ascending powers of x.

14 Use the binomial theorem with $(1-0.01)^4$ to approximate 0.99^4 to four decimal places, using the first three terms.